

Sugano Organic — Biomass CHP Pilot

Technical Requirements & Proposal · Phase 1 Functional Prototype

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Purpose

This document outlines the technical requirements and investment case for installing a small biomass combined heat and power (CHP) system at Sugano Organic, powered entirely by Sugi trees from Tokuo's own mountain forest.

Sugano Organic becomes **Phase 1 of the Mitsue Project's energy programme** — a real, working prototype demonstrating that a rural home and workshop can be fully powered and heated by the forest around it. Every product made at Sugano Organic thereafter carries that proof.

Site Overview

Item	Detail
Location	Mitsue Village, Nara Prefecture
Electricity need	~10 kW peak (house + workshop)
Heating need	~23 kW (winter, house + factory)
Fuel source	Tokuo's own 28.05 ha Sugi mountain forest
Grid connection	Retain as backup / export surplus

System Design

Core: 10 kW Wood Gasifier + Generator

A wood gasifier converts Sugi logs and chips into combustible gas (syngas), which drives a generator producing electricity. Waste heat from the engine is captured for space and water heating.

Parameter	Value
Electrical output	10 kW
Thermal input required	45.5 kW
System efficiency (electrical)	22%
Wood consumption	10.9 kg/hour
Daily consumption (8h operation)	87 kg/day
Annual consumption (250 days)	21.8 tonnes/year

Parameter	Value
Annual electricity generated	20,000 kWh/year

Heat Recovery (CHP)

The engine produces far more heat than electricity. Capturing this waste heat raises overall system efficiency from 22% to 84%.

Parameter	Value
Waste heat available	35.5 kW
Recoverable (80% efficiency)	28.4 kW
House + workshop heating need	~23 kW ✓
Heating saving vs kerosene	¥340,000/year
Overall CHP efficiency	84%

CHP Components

- Water-cooled heat exchanger jacket on engine
- 500–1,000 L insulated hot water storage tank
- Piping and insulation (house + workshop circuit)
- Radiators or underfloor heating connections
- Control and integration unit

Fuel Supply: Tokuo's Forest

Parameter	Value
Forest area owned	28.05 ha
Rotation cycle	30 years
Annual sustainable harvest	0.94 ha/year → 5.6 tonnes/year
Initial thinning (one-time)	~196 tonnes (35% of overgrown stock)
Fuel from initial thinning alone	~9 years at 10 kW consumption

Key point: Mitsue's Sugi plantations are heavily overgrown and actively need thinning for forest health. The initial clearance is not a cost — it is forest restoration work that also produces the first 9 years of fuel for free.

Long-term (after year 9), a supplementary ~16 tonnes/year can be sourced from Mitsue Village community forest — a small volume easily arranged through Mitsue Kanko or the village forest cooperative.

Wood Processing Requirements

The system requires air-dried wood chips or split logs (15–20% moisture content).

Equipment needed	Note
Chainsaw + log splitter	For initial thinning — may already be owned
Wood chipper (if chips preferred)	Rental or shared with Mitsue Kanko
Covered wood drying storage	~50 m² shed, 3–6 months drying time

Mitsue Kanko can potentially coordinate the annual thinning and processing as a service, supporting multiple village landowners over time.

Investment & Financial Summary

System Costs (Japan market, 2026)

Item	Low estimate	High estimate
10 kW gasifier + generator unit	¥2,800,000	¥5,500,000
Installation (~25% of unit cost)	¥700,000	¥1,375,000
CHP heat recovery system	¥1,500,000	¥3,500,000
Total investment	¥5,000,000	¥10,375,000
Approximate USD	\$33,000	\$69,000

Annual Savings

Item	Value
Grid electricity avoided	¥540,000/year
Kerosene heating avoided	¥340,000/year
Combined annual saving	¥880,000/year

Payback Period

Scenario	Payback
Without subsidies	5–11 years
With 50% subsidy (realistic)	3–6 years
CO ₂ avoided	9.0 tonnes/year

Subsidy & Funding Pathways

The Mitsue Project NGO (currently being established) will actively pursue the following grants on behalf of Tokuo:

Programme	Body	Typical support
Rural biomass energy subsidy	METI / 経済産業省	Up to 50% of system cost
Agricultural biomass support	MAFF / 農林水産省	Equipment grants for wood energy
Forest thinning subsidy	Forestry Agency / 林野庁	¥50,000–¥80,000/ha thinned
Carbon credit income	J-Credit scheme	¥2,000–¥5,000/tonne CO ₂

The NGO acts as the **financial and administrative tunnel** — handling applications, coordinating with government, and channelling grant funding directly to the site.

Prototype Value for the Mitsue Project

Tokuo's site is the ideal Phase 1 prototype because:

1. **He already owns the fuel** — no land acquisition, no supply chain to build
2. **Small, manageable scale** — 10 kW is technically simple and low-risk
3. **Dual use** — powers both a home and a creative workshop
4. **Strong story** — natural clothing + nature sounds, powered by the forest. Authentic and communicable worldwide
5. **Replicable** — dozens of Mitsue households and small businesses face the same situation

Data from this installation (fuel consumption, uptime, costs, CO₂) feeds directly into the larger Mitsue Project energy planning and grant applications.

Proposed Next Steps

1. **Site visit** — walk the forest together, assess thinning priorities and wood storage location
2. **Contractor shortlist** — identify 2–3 Japanese gasifier suppliers for quotes (e.g. Ankur Scientific, local fabricators)
3. **NGO grant scan** — Rob's team identifies applicable subsidy programmes and eligibility
4. **Feasibility sign-off** — confirm grid connection requirements with local utility
5. **Phase 1 decision** — agree scope, budget, and start date

Summary

Tokuo's own Sugi mountain — the forest his family has tended for generations — is ready to power his home and workshop completely, cleanly, and independently.

A 10 kW biomass CHP system costing ¥5M–¥10M, with a realistic payback of 3–6 years after subsidies, producing 20,000 kWh of electricity and 28 kW of winter heat from wood that would otherwise go unthinned.

先人が植えた森が、今、村を支える力になる。 *The forest our ancestors planted — the power that sustains the village they built.*